REUBEN NEWTON ADDISON, PhD

Clinical Research Scientist | Biomechanics | Clinical Data Science

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PROFESSIONAL PROFILE

Clinical research scientist and data scientist with a foundation in motor behavior, neurological research, and human-subjects study design. Lead the Motor Control Lab at DePauw University and bring experience in Parkinson's disease, longitudinal neuroimaging analysis, Bayesian methods, and biomedical machine learning. Develop reproducible analyses and research software that connect clinical questions to interpretable evidence.

RESEARCH AND TECHNICAL EXPERTISE

Clinical research: Human-subjects and observational studies; protocol development; IRB-governed research; clinical measurement; evidence synthesis; treatment-response analysis.

Statistics: Bayesian inference; multilevel and mixed-effects models; longitudinal and repeated-measures analysis; experimental design; sensitivity analysis; simulation.

Data science: Python, R, SQL, MATLAB, PyMC, PyTorch, scikit-learn, XGBoost; predictive modeling; CNNs and Transformers; reproducible workflows.

Movement and biomedical data: Motion capture, kinematics, behavioral outcomes, EEG, ECG, fMRI, and TMS.

PROFESSIONAL EXPERIENCE

Assistant Professor of Kinesiology and Director of the Motor Control Lab

DePauw University, Greencastle, IN | Aug 2023 - Present

- Lead IRB-governed research on neurological function, Parkinson's disease, and movement outcomes, from study design and data collection through statistical analysis and manuscript development.
- Apply Bayesian and mixed-effects models to repeated observations and treatment-response questions; interpret findings with attention to uncertainty and participant variability.
- Collaborate on research design and grant development, mentor undergraduate researchers, and teach biomechanics, motor control, and applied quantitative methods.

Data Science Intern

Chinese American Biopharmaceutical Society, Remote | Jun 2026 - Sep 2026

- Built OpenTrial, a Bayesian clinical-trial design engine connecting clinical evidence synthesis with prior construction, power and assurance analysis, and simulation.
- Integrated ClinicalTrials.gov, PubMed, openFDA, and DailyMed evidence into source-cited reporting workflows and reviewed clinical and regulatory materials.

Postdoctoral Research Fellow

MGH Institute of Health Professions, Brain Recovery Lab | Jun 2022 - Jun 2023

- Analyzed longitudinal fMRI and TMS data from NIH-funded neurological studies using multilevel models and reproducible preprocessing workflows.
- Prepared statistical results for clinical collaborators and NIH-facing discussions; mentored undergraduate researchers.

EARLIER RESEARCH EXPERIENCE

Graduate Research and Teaching Assistant

Louisiana State University | *Aug 2017 - May 2022*

- Designed and analyzed studies of visuomotor adaptation, motor learning, and Parkinson's disease, including dietary-intervention research and movement-performance outcomes.
- Conducted EEG and behavioral research, authored peer-reviewed and conference work, and taught and guest lectured in movement science and quantitative methods.

Mitacs Accelerate Research Intern

Memorial University, Faculty of Medicine | *Dec 2015 - Jul 2016*

- Evaluated a tele-ultrasound system in a clinical feasibility study supported by $10,000 CAD in competitive funding; co-authored a peer-reviewed publication.

SELECTED RESEARCH AND DATA SCIENCE PROJECTS

OpenTrial

Bayesian clinical trial design | Python | 2026

Developed source-cited design reports with evidence-derived priors, power and assurance analysis, prior sensitivity, and simulation-calibrated group-sequential designs. Research and learning engine; not validated for clinical use. github.com/reubebe/OpenTrial

KinetiScan

Markerless movement analysis | Python, PyTorch, ONNX | Ongoing

Developing video-based 3D kinematics and gait analysis with monocular and multi-camera pathways. Separate kinematics-engine evaluation from end-to-end camera validation, which remains to be established.

ECG Signal Quality Classification

Biomedical machine learning | PyTorch, PTB-XL | 2026

Compared five statistical and neural classifiers, including a CNN-Transformer ensemble, with threshold optimization, calibration assessment, and saliency and lead-level interpretation. github.com/reubebe/ECG-signal-quality-detection

EDUCATION

MS in Analytics | Georgia Institute of Technology | May 2026 | Computational Data Analytics Track

PhD in Kinesiology | Louisiana State University | May 2022 | Motor Behavior

MS in Kinesiology | Memorial University of Newfoundland | May 2017

BS in Psychology | University of Ghana | May 2012

SELECTED SCHOLARSHIP

Addison, R. N., & Van Gemmert, A. W. (2023). Bilateral transfer of a visuomotor task in different workspace configurations. Journal of Motor Behavior. doi:10.1080/00222895.2023.2293002.

Smith, A., Addison, R., et al. (2018). Remote mentoring of point-of-care ultrasound skills to inexperienced operators using multiple telemedicine platforms: Is a cell phone good enough? Journal of Ultrasound in Medicine, 37(11), 2517-2525. doi:10.1002/jum.14609.